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Gas Reservoirs Predict Future Galaxy Growth Beyond Halo Assembly History in CAMELS

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ABSTRACT

Does a galaxy's gas content tell us anything about its future growth once its dark-matter halo history is already known? We test this question in the CAMELS CV suite using 27 fixed-parameter realizations of the IllustrisTNG and SIMBA galaxy-formation models. For each galaxy, we compare three predictor families — internal galaxy properties, halo/assembly-history properties, and environment — as predictors of future stellar growth, quenching, and halo growth from $z \simeq 0.77$ to $z = 0$. We measure how much predictive power each family adds beyond increasingly strict halo controls, ending with a 13-feature assembly-history baseline.

In TNG, gas reservoir information predicts future stellar growth beyond the full assembly-history control at low-to-intermediate stellar mass, up to $\log M_*/M_\odot \simeq 10.55$. This is not simply a current-star-formation effect: the gas signal survives controls for stellar mass, SFR, and sSFR. It is also not a gas-use-efficiency effect: gas amount remains informative, while depletion time adds no comparable information once gas amount is included. Halo assembly history strongly predicts the gas reservoir itself ($R^2 \simeq 0.8$), but the residual gas component still predicts future growth. Thus the gas reservoir is shaped by halo history but not exhausted by it.

Near $\log M_*/M_\odot \simeq 10.55$, the predictive structure changes. Black-hole/quenching state, traced by catalog-level black-hole proxies rather than direct feedback-energy measurements, becomes more informative, and the high-mass population is largely quenched. The full assembly-history (L3) control is available only for TNG; because SubLink-equivalent merger trees are unavailable for SIMBA, we compare the two models at a matched static-halo (L1) control, where they express residual baryonic information in different channels — stellar growth in TNG and quenching in SIMBA — with no SIMBA L3-controlled claim. We caution that the low-mass edge is resolution/floor-sensitive rather than a sharp physical threshold, and that the environment result is limited to the CAMELS CV volume and features. Overall, the results are consistent with a fuel-limited to quenching-limited picture of future galaxy growth and motivate an observational test: gas-rich galaxies should grow preferentially at fixed stellar mass and assembly proxies in surveys such as xGASS / xCOLD GASS.

Keywords: galaxies: evolution — galaxies: statistics — methods: statistical — dark matter haloes — galaxies: star formation — galaxies: quenching

1. INTRODUCTION

A central question in galaxy formation is which properties best predict how a galaxy will grow or quench over the coming few gigayears. Three families of properties are routinely discussed: the internal baryonic state of the galaxy (gas content, star-formation rate, stellar mass), the properties of its host dark matter halo (mass, concentration, spin, formation history), and the large-scale environment (local density, group richness, filamentary structure). A substantial literature has addressed each family in isolation (e.g., R. Davé et al. 2012; A. R. Wetzel et al. 2013; P. Behroozi et al. 2019; B. P. Moster et al. 2013; Y. Peng et al. 2015), and a growing body of work connects galaxies to their dark matter halos across cosmic time (R. H. Wechsler & J. L. Tinker 2018; L. Gao et al. 2005; R. H. Wechsler et al. 2006; Y.-Y. Mao et al. 2018), but systematic comparisons under the same sample, time horizon, and control structure are less common.

The difficulty is that these families are not independent. Halo mass correlates with environment. Gas content correlates with halo assembly history through the rate at which gas is accreted — via cold streams at high redshift and hot-halo cooling at later times (A. Dekel et al. 2009) — and feedback-driven outflows are recycled (N. Bouché et al. 2010; S. J. Lilly et al. 2013). Stellar mass — part of any internal description — is itself a record of past growth, and therefore implicitly encodes assembly history. Correlations between a property family and a future outcome can therefore reflect the gravitational context rather than baryonic processes specific to that family. The standard approach of correlating present-day properties with future states — whether via classical statistics or machine-learning feature ranking (e.g., A. F. L. Bluck et al. 2023; J. J. Davies et al. 2020; H. Teimoorinia et al. 2016; C. C. Lovell et al. 2019; J. Matthee & J. Schaye 2019) — does not cleanly separate these contributions.

A sharper question is: after controlling for what the halo structural and assembly history already predicts, does a galaxy's internal baryonic state carry additional information about its future? And does the answer depend on stellar mass? Several results suggest it might. Observations and simulations show that galaxy growth at intermediate stellar mass ($\log M_* \sim 9.5$–10.5) is particularly sensitive to gas supply and feedback (R. Davé et al. 2012; A. Pillepich et al. 2018a; A. Saintonge et al. 2017), while at low mass, growth tracks the large-scale structure and tidal field, and at high mass, AGN feedback and the halo potential increasingly dominate quenching (D. J. Croton et al. 2006; R. Weinberger et al. 2018; Y. Peng et al. 2015). If the coupling between baryonic state and future growth is genuinely mass-regime-specific, then a global comparison of predictor families will miss the structure.

We address this directly using the CAMELS Cosmic Variance (CV) suite (F. Villaescusa-Navarro et al. 2021), which provides 27 independent realizations each of IllustrisTNG (R. Weinberger et al. 2017; A. Pillepich et al. 2018b) and SIMBA (R. Davé et al. 2019) at fixed astrophysical parameters. The statistical independence of the 27 realizations allows bootstrap confidence intervals on predictive scores to be evaluated at population level rather than for individual realizations. We construct three predictor families from early-epoch ($z \approx 0.77$) galaxy properties and ask how well each predicts three outcomes at $z = 0$ (stellar growth, quenching, and halo growth), both in the whole sample and as a function of stellar mass. The central diagnostic is a marginal score: the increment in cross-validated predictive performance when a feature family is added to a halo assembly-history control, evaluated by shared bootstrap. This directly measures whether a feature family carries predictive information beyond the halo's gravitational record.

The key developments are as follows. No feature family leads consistently across all targets and mass regimes: the 3×3 performance matrix shows statistical ties in both simulation families. Spatial matching inflates the assembly-history baseline $3.7\times$ relative to merger-tree matching, a confound that must be accounted for before cross-family assembly-control comparisons. At low-to-intermediate stellar mass — up to $\log M_*/M_\odot \simeq 10.55$ — internal gas-reservoir information retains predictive power for future stellar growth beyond the full 13-feature assembly-history control. This residual is carried by the gas *amount*: it survives controls for stellar mass and for the current (specific) star-formation rate, and it is not a gas-use-efficiency (depletion-time) effect. The reservoir is itself strongly predicted by halo assembly history, yet a residual gas component continues to predict growth — so the gas reservoir is shaped by halo assembly history but not exhausted by it. Near $\log M_*/M_\odot \simeq 10.55$ the predictive structure changes: black-hole and quenching state — traced by catalog-level black-hole proxies rather than direct feedback-energy measurements — becomes the more informative channel, and the high-mass population is largely quenched. The full L3 assembly-history control is available only for TNG; because SubLink-equivalent merger trees are unavailable for SIMBA, we compare the two feedback models at a matched static-halo (L1) control, where they express this residual baryonic information in different channels — stellar growth in TNG and quenching in SIMBA — and we make no SIMBA L3-controlled claim. The lower edge of this regime is set by resolution and catalog-floor effects rather than a sharp physical threshold; results are otherwise stable across all 27 CAMELS CV realizations and to the choice of halo structural variable.

2. DATA AND METHODS

2.1. *CAMELS CV Suite*

We use the Cosmic Variance (CV) sub-suite of CAMELS, comprising 27 independent realizations each of IllustrisTNG and SIMBA at fixed cosmological and astrophysical parameters (F. Villaescusa-Navarro et al. 2021). Each box is $25\,h^{-1}\,\mathrm{Mpc}$ with 256^3 dark matter particles; we use snapshots 066 and 090 ($z = 0.7747$ and $z = 0$). We restrict to *central* galaxies with $\log M_*(z = 0.77) \geq 9.0$; after matching the TNG SubLink sample contains $n = 7{,}362$ galaxies and the SIMBA spatial sample $n = 4{,}203$.

2.2. *Predictor Feature Families*

We define three predictor families, each evaluated independently using only features from within that family.

All features are measured at the early epoch ($z = 0.77$). The three families are evaluated independently so that their marginal contributions can be assessed cleanly; no family has access to features of any other family when scored.

Table 1. Predictor feature families.

Family	$N_{\rm feat}$	Features	Source
Environmental	4	$\log M_{\rm halo}$, $\log N_{\rm subs}$, $\log \rho_{\rm local}$, $\log n_{\rm 5Mpc}$	FoF halo + neighbor count
Internal	6	$\log M_*$, $\log M_{\rm gas}$, $\log {\rm SFR}$, $\log {\rm sSFR}$, $\log R_*$, metallicity	Subhalo catalog
Halo structural	5	$\log M_{\rm sub}$, $V_{\rm max}$, $\log R_{\rm sub}$, spin, $V_{\rm max}/V_{200}$	Subhalo + FoF

2.3. *Targets*

Three outcomes are predicted at $z = 0$: stellar growth ($\Delta\log M_* = \log M_*(z = 0) - \log M_*(z = 0.77)$, Ridge R^2, 5-fold CV); binary quenching ($\text{sSFR}(z = 0) < 10^{-11}\,\text{yr}^{-1}$, logistic Ridge AUC, 5-fold CV); and halo growth ($\Delta \log M_{\rm sub}$, Ridge R^2). Hyperparameters are tuned by inner cross-validation on the training fold.

2.4. *Matching Methods*

Tree-based SubLink matching (V. Rodriguez-Gomez et al. 2015), available for TNG only, links each early-epoch galaxy to its $z = 0$ descendant via merger trees, introducing no structural selection bias; it is used for all assembly-history control analyses. Spatial matching — nearest late-epoch galaxy within $3\times$ the early-epoch half-mass radius — preferentially selects structurally stable systems, inflating the assembly-history baseline (Section 3.2). It is used for SIMBA, where SubLink-equivalent trees are unavailable.

2.5. *Significance Rule*

A feature family is declared to lead over another if the bootstrap 95% CI lower bound of their score gap exceeds 0.02 (R^2 or AUC); otherwise the result is declared a tie. The threshold of 0.02 was fixed before interpretation and is not tuned to produce a particular result. It represents the minimum practically meaningful gap relative to the observed bootstrap noise structure in the 27-simulation CV suite: gaps below 0.02 are within the regime where small changes in simulation selection or minor specification differences routinely flip the ordering sign, making them uninformative as science claims. The main results of this paper — the upper-bounded intermediate-mass regime, the broadly flat paired gap, the carrier and ablation decompositions — are continuous statistics evaluated with 95% bootstrap CIs and do not depend on this threshold. The TIE/WINNER declarations in Sections 3.1–3.3 do depend on it; those conclusions are qualitatively stable to reasonable threshold variation (at 0.01, the mid-mass growth WINNER verdict is unchanged; at 0.05, only the strongest gaps survive, but the mass-regime patterns are preserved).

2.6. *Three-Layer Assembly-History Controls*

To assess whether predictor family signals persist beyond what the halo structural and assembly history already predicts, we apply a three-layer control procedure to TNG SubLink.

L1 (static halo structure) uses three features describing where the halo sits at $z = 0.77$: $\log M_{\rm halo}$, $\log \rho_{\rm local}$, and $\log M_{\rm sub}$. L2 augments L1 with $\Delta \log M_{\rm halo}$ over 4 and 8 SubLink steps ($\approx$ 1–2 Gyr lookback) and halo formation snapshot. L3 extends L2 with $\Delta \log M_{\rm halo}$ over 12 and 16 steps ($\approx$ 3–4 Gyr lookback), peak mass ratio, half-mass assembly snapshot, merger count, major-merger count (mass ratio ≥ 0.25), and last major-merger snapshot, for a total of 13 assembly-history features.

Three control statistics are computed for each predictor family. The *marginal* R^2/AUC is the increment in cross-validated score when the family is added to the assembly-history control; 95% CIs come from a shared bootstrap ($n_{\rm boot} = 1000$). *Residualization* OLS-regresses each predictor feature column against the assembly-history control set and replaces it with the OLS residual before the predictive model is fit (it is the feature columns, not the target, that are residualized). *Retention* is the ratio of residualized to original score, with the AUC analog computed on the $(x - 0.5)$ excess.

2.7. *Robustness Tests*

Two robustness checks are applied throughout:

1. *Simulation jackknife*: leave one of 27 CV realizations out; report the fraction $n_{\rm agree}/27$ that preserve the full-sample ordering.

2. *Permutation importance*: shuffle each feature column; measure the mean drop in CV R^2 over $n_{\rm boot} = 500$ draws.

3. RESULTS

3.1. *No Feature Family Leads Consistently Across Targets*

The full 3×3 score matrix is in Appendix D (Table 8). In both TNG SubLink and SIMBA spatial, every cell is a tie: no feature family leads across all three targets, and every pairwise gap has a 95% CI spanning or approaching zero.

The tie is robust across both simulation families: TNG and SIMBA agree on winner structure despite differing in overall predictability level. Within-target ordering is target-dependent (growth favors internal at mid mass; quenching favors internal in both), motivating the mass-regime decomposition below. Cross-family quantitative differences are partly a matching artifact (Section 3.2) and are not interpreted as clean physics signals. Halo growth ($\Delta \log M_{\rm sub}$) is near-unpredictable by any family at $z \approx 0.77$ ($R^2 \leq 0.015$ across all three families in TNG; Table 8). This bound is for the whole-sample structural families; even the richer 13-feature L3 assembly-history context predicts whole-sample halo growth at only $R^2 \approx 0.029$, and in the intermediate window every family remains $\lesssim 0.016$ (internal $+0.009$, halo $+0.016$, environment $+0.000$) — an order of magnitude below stellar growth, where the internal family alone reaches $R^2 \sim 0.2$–0.3 in the same window. This indicates that dark matter accretion between $z = 0.77$ and $z = 0$ is driven by large-scale density fluctuations not encoded in the snapshot properties of individual galaxies. This contrast is itself informative: late dark-matter accretion is not well encoded in individual $z \simeq 0.77$ subhalo, galaxy, or environment properties in CAMELS CV, whereas baryonic stellar growth retains stronger predictive structure — a galaxy's baryonic future is more foreseeable than its gravitational future at fixed epoch.

3.2. *Matching Method Confound*

The assembly-history baseline — the R^2 achievable from static structural position alone — differs dramatically by matching method (Table 9 in Appendix D): SubLink gives $R^2 = 0.059\,[0.051, 0.072]$; spatial matching gives $R^2 = 0.220$ (TNG) and $R^2 = 0.273$ (SIMBA).

Spatial matching selects galaxies that remain structurally stable between $z = 0.77$ and $z = 0$, inflating all predictor scores and the assembly-history baseline 3.7×. The similarity between TNG spatial and SIMBA spatial baselines is partly a matching artifact, not a physics signal. **All assembly-history control analyses in this paper use TNG SubLink exclusively.** The mechanistic findings of Sections 3.4–3.7 do not extend to SIMBA without SubLink-equivalent merger trees.

3.3. *Mass-Regime Splits: The Whole-Sample Tie is Misleading*

The whole-sample tie conceals opposite orderings at different stellar masses (Table 7, Appendix D). Mid mass ($9.5 \leq \log M_* < 10.5$) is the only regime with a clear growth winner: internal $R^2 = 0.307$ versus halo $= 0.235$, gap $= +0.072$ (bootstrap 95% CI lower bound > 0.02; see Table 2 for the assembly-history-controlled lower bound, $+0.052$–$+0.089$ across 13 overlapping windows). At low mass, internal growth is near noise while environmental and halo features lead; at high mass, the quenching ordering partially inverts. The whole-sample tie is a population mixture of three regimes with different orderings.

At low and high mass, predictor advantages largely disappear under assembly-history controls: the L1 halo structural baseline accounts for most of the environmental/halo low-mass growth advantage, and L1+L2 absorbs the remainder. High-mass quenching is entirely structural-position-carried at all control layers — no feature family adds marginal signal beyond where the galaxy sits. Sections 3.4–3.7 concern mid-mass growth, where the internal family retains signal under progressively stronger assembly-history controls.

3.4. *An Upper-Bounded Intermediate-Mass Internal Signal*

To characterize where internal state's advantage begins and ends, we apply a sliding 0.5-dex window (step 0.1 dex, $n_{\rm boot} = 500$) across the full mass range and compute L1 marginal R^2 at each centre (Figure 2; selected window positions in Table 10, Appendix D). Because the windows are 0.5 dex wide and stepped by only 0.1 dex, adjacent windows overlap heavily and are strongly correlated; the window counts below describe the contiguous *extent* of significance in the scan, not independent tests.

In the unfiltered catalog the internal marginal is negative or noisy below ∼9.55 dex and first enters significance at +0.149*; halo enters three windows earlier (9.25 dex), so structural accretion carries growth information at lower mass before internal state does. This lower boundary is *not* a physical threshold. It is a floor-encoding / resolution-limited recovery edge: below ∼9.55 dex a small tail of unresolved objects whose gas feature sits at the catalog floor destabilizes the fit, and repairing that encoding — either by removing the tail or by winsorizing the floor-encoded feature *without removing any object* — restores a positive gas marginal below the nominal edge (+0.061 after removal and +0.057 after winsorization, versus ∼ 0 unfiltered). We therefore treat 9.55 dex as the onset of *recoverable* internal signal in the unfiltered catalog, not as a sharp physical onset. The signal rises to a peak of +0.245* at 10.25 dex — the highest value in the scan — with both families significant and internal leading by a growing margin toward the centre. The upper close, by contrast, is physically meaningful: internal remains significant through 10.65 dex, drops to ns at 10.75 dex, and both families collapse together above that mass. The unfiltered interval over which the internal L1

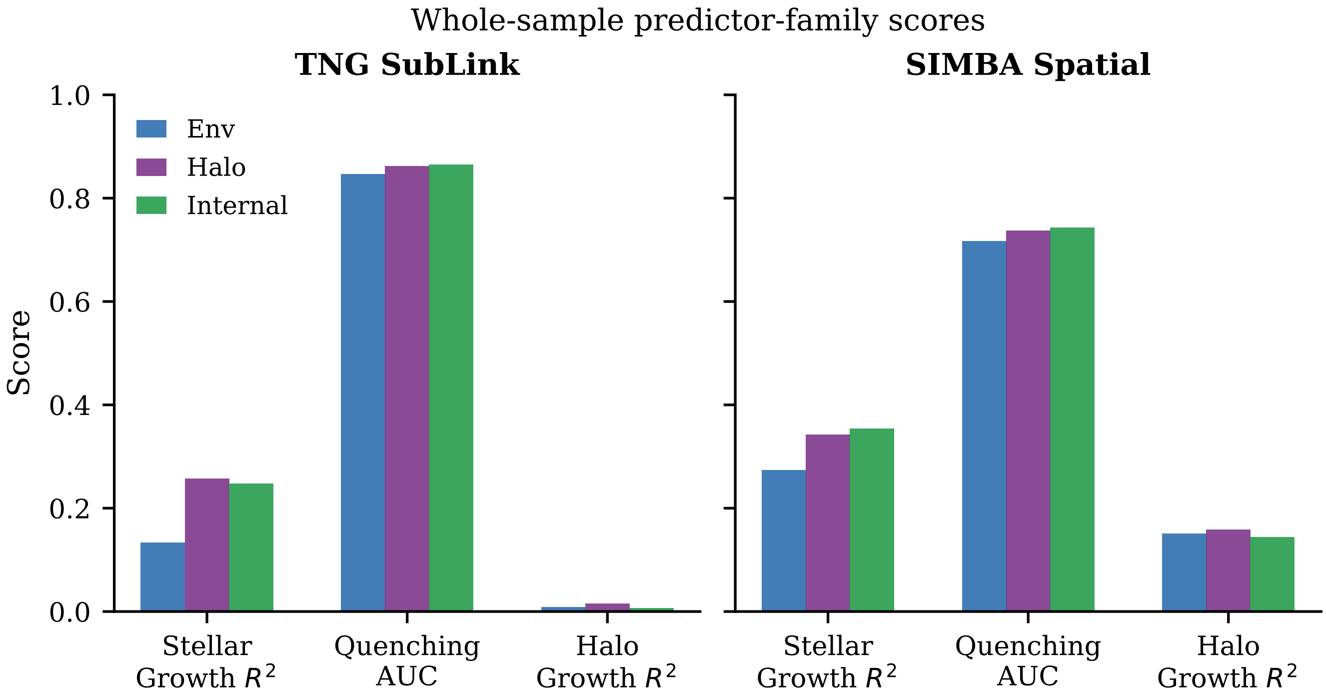

Figure 1. Full 3×3 score matrices for TNG SubLink (left) and SIMBA spatial (right). Color encodes class score (R^2 or AUC). All cells are statistical ties in both simulation families. The whole-sample tie is the surface pattern; the mass-regime structure revealed by sliding-window analysis is the main result (Figure 2).

marginal is individually significant is thus 9.55–10.65 dex (13 overlapping windows); under the full 13-feature L3 baseline it is 9.55–10.55 dex (Section 3.5).

3.5. *The Signal Survives Full Gravity Control*

The L1 interval above uses only static structural position as control. We test whether the signal survives a 13-feature L3 assembly-history control comprising: L1 features, four accretion-rate windows (1–4 Gyr lookback), halo formation snapshot, peak mass ratio, half-mass assembly snapshot, merger count, major-merger count, and last major-merger timing.

L3 assembly-history baseline across this interval averages $R^2_{\rm geom} \approx 0.16$–$0.18$, compared to L1's ≈ 0.004 at mid mass — the full assembly history prescription is far stronger than static position. Nonetheless, the internal signal survives under L3 (Figure 2, lower panel; selected positions in Table 11, Appendix D).

Under L3, the interval over which internal $R^2_{\rm marg}$ is individually significant spans **9.55–10.55 dex** (11 overlapping windows); as above, its 9.55-dex lower bound is recovery-limited, while the 10.55-dex upper bound is the physical edge of the regime. The onset is unchanged. The close moves one window inward (10.55 vs. 10.65 at L1). The amplitude at the peak window is $\approx 65\%$ of the L1 peak: full assembly history absorbs $\sim 35\%$ of the internal advantage within this interval, but $\sim 65\%$ survives.

Halo's significant interval collapses dramatically under L3: 14 significant windows at L1 reduce to 3 non-contiguous windows (9.85, 10.15, 10.35 dex) at L3. Most of halo's growth predictability across the mass range is accounted for by the 13 assembly-history features. Internal's significant interval — narrower at L1 than halo's — is the more robustly baryonic of the two.

Env carries no significant marginal in any L3 window; its growth information is entirely gravity-proxied.

3.6. *Internal Leads Halo Across the Entire Regime*

Figure 3 shows the shared-bootstrap paired gap $\delta = R^2(\text{L3} + \text{internal}) - R^2(\text{L3} + \text{halo})$ at each sliding window centre. The assembly-history baseline cancels algebraically in this test, giving a direct internal-vs-halo comparison under full gravitational control; this algebraic cancellation makes the paired gap the paper's most statistically robust diagnostic, immune to uncertainty in the absolute level of the baseline.

The gap is significant at every window from 9.55 to 10.75 dex — 13 overlapping windows. The geometry cancellation extends this significant range two windows beyond the 9.55–10.55 dex internal interval: internal $R^2_{\rm marg}$ becomes ns at 10.65 dex, but the paired gap persists to 10.75 dex because the baseline cancels al-

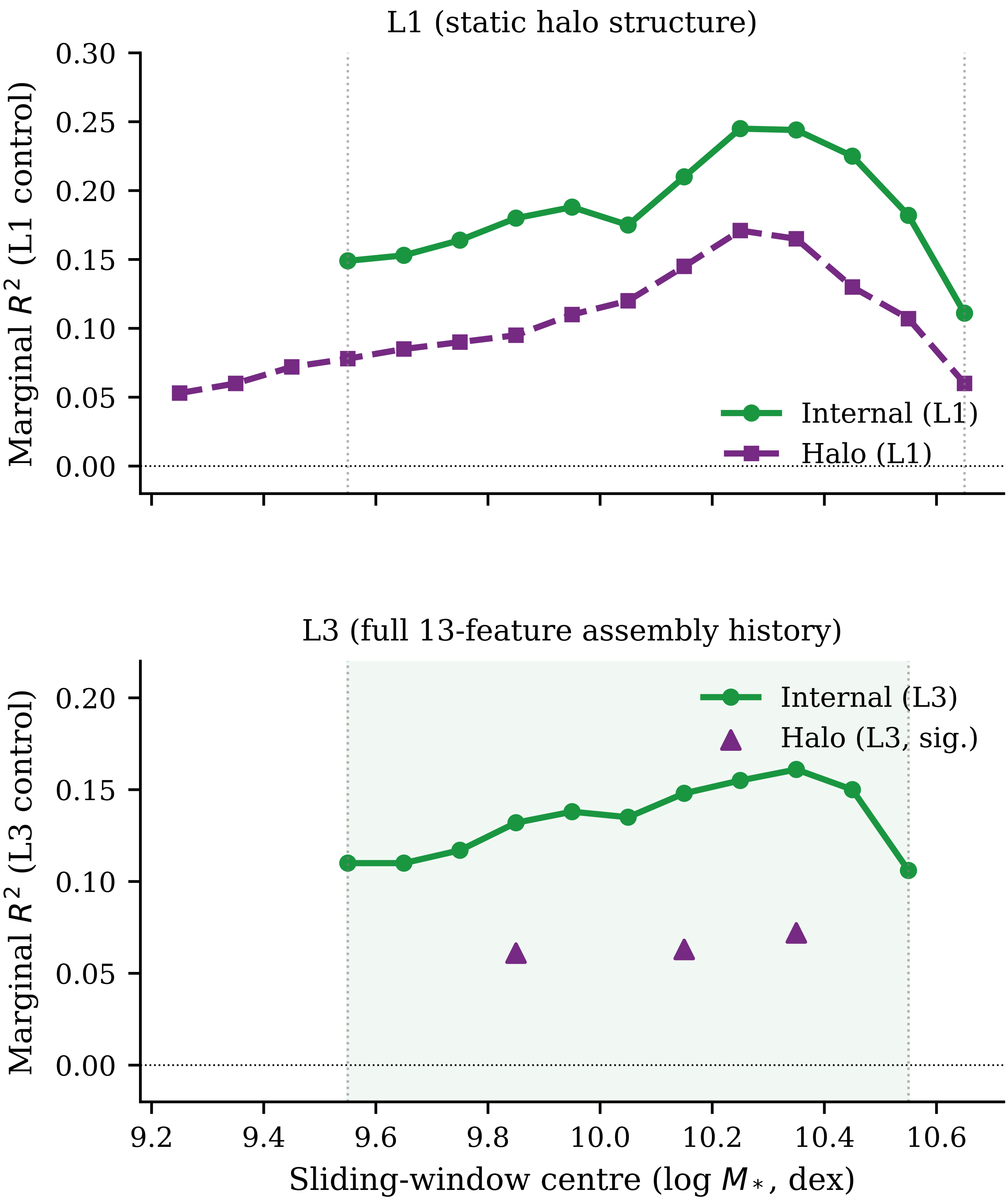

Figure 2. Intermediate-mass growth phase diagram. Internal (solid) and halo L1 marginal R^2 (upper) and L3 marginal R^2 (lower) versus sliding-window centre (0.5-dex windows, step 0.1 dex). L1 significant interval: 9.55–10.65 dex (13 windows). L3 significant interval (green shading): 9.55–10.55 dex (11 overlapping significant windows), where internal $R^2_{\rm marg}$ is individually significant. L3 halo collapses to 3 non-contiguous windows (triangles); the halo L3 markers are shown as disconnected triangles, with no connecting line, because the significant halo intervals are non-contiguous. Vertical dotted lines mark the L3 significant interval (9.55–10.55 dex).

Table 2. L3 paired gap (internal − halo) by window centre.

Centre (dex)	n	δ	95% CI	Sig
9.25–9.45	⋯	< 0	spans zero	ns
9.55	2,393	+0.071	[+0.049, +0.095]	*
9.65	2,213	+0.067	[+0.042, +0.085]	*
9.75	2,023	+0.066	[+0.042, +0.089]	*
9.85	1,908	+0.071	[+0.047, +0.094]	*
9.95	1,747	+0.072	[+0.047, +0.091]	*
10.05	1,566	+0.071	[+0.043, +0.093]	*
10.15	1,459	+0.067	[+0.046, +0.084]	*
10.25	1,407	+0.065	[+0.049, +0.087]	*
10.35	1,337	+0.089	[+0.051, +0.101]	*
10.45	1,228	+0.071	[+0.044, +0.094]	*
10.55	1,158	+0.052	[+0.030, +0.085]	*
10.65	1,026	+0.026	[+0.016, +0.071]	*
10.75	836	+0.016	[+0.004, +0.061]	*
≥ 10.85	⋯	ns	⋯	

NOTE—$*$ denotes 95% CI entirely above zero.

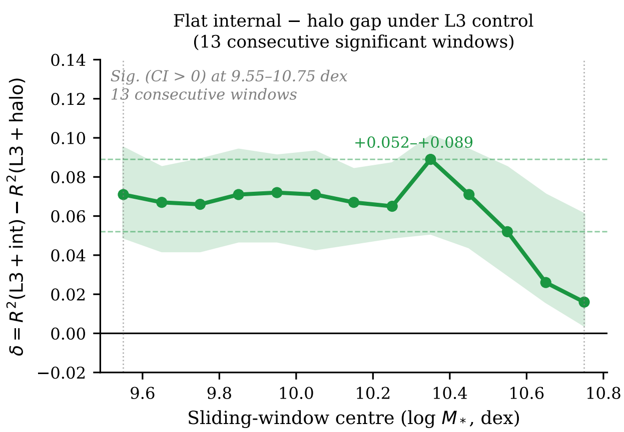

Figure 3. Flat internal–halo paired gap across the regime. L3 shared-bootstrap paired gap $\delta = R^2(\mathrm{L3}+\mathrm{internal}) - R^2(\mathrm{L3}+\mathrm{halo})$ versus window centre with 95% CI shading. The paired gap is significant over 9.55–10.75 dex (13 overlapping windows), extending two windows beyond the 9.55–10.55 dex internal interval, because the geometry cancellation in the paired test removes baseline noise. Gap ranges +0.052–+0.089 with no strong mass dependence.

gebraically in the shared-bootstrap paired test. As for the marginal interval, the 9.55-dex lower bound of this gap interval reflects catalog recoverability rather than a physical threshold; only the upper bound marks a physical transition. Across that upper edge the population changes character: between the 9.55–10.55 dex interval and the 10.55–10.75 dex band the median specific star-formation rate drops from $\log \mathrm{sSFR} = -9.63$ to -10.96, the quenched fraction rises from 0.035 to 0.498, and the median black-hole mass increases by 1.51 dex — the 10.55-dex boundary coincides with the onset of black-hole–associated quenching, which is why it, unlike the lower edge, is a physical transition. Appendix Figure 14 shows these population profiles as a function of stellar mass.

The gap is also broadly flat, with no systematic trend across mass: over 1.2 dex of stellar mass, δ ranges +0.052 to +0.089 without strong peaking or an edge effect. Internal's advantage over halo is a stable property of the entire intermediate-mass regime, not an artefact of a single peak mass. The simulation jackknife confirms this: the ordering internal > halo under L3 is preserved at all 9.55–10.55 dex windows in 27/27 leave-one-out configurations.

3.7. *Mechanistic Decomposition*

3.7.1. *Gas mass is the dominant surviving internal carrier*

To identify which features within the internal family carry the residual signal, we apply permutation importance to L3-residualized columns at the mid-mass regime, with each feature regressed against the 13 assembly-history variables and replaced by its OLS residual before shuffling (Table 3).

The result is unambiguous: gas mass is the sole internal feature retaining significant importance after full gravity residualization. All other internal features — SFR, sSFR, stellar mass, metallicity, half-mass radius — are absorbed by the L3 assembly-history. Stellar-to-halo fraction f_* is the most gravity-resistant feature in the entire analysis, retaining 84% of its raw permutation importance after L3 residualization.

3.7.2. *Assembly timing, not mergers, is the main absorbing gravitational channel*

From the weak-L1 control (static halo position only) to the full 13-feature L3 baseline, the internal-family marginal for mid-mass growth falls from 0.308 to 0.135, so 0.173 in R^2 is absorbed by assembly history. Peak mass ratio and half-mass assembly snapshot together account for 19% of the absorbed internal marginal. Late-lookback accretion (4–8 Gyr) accounts for a further 10%. Merger counts recover zero internal signal: the residual internal signal in the mid-mass regime is entirely merger-count-independent. Individually, the tested components each recover only a minority of the absorbed internal signal at mid-mass (timing 19%, long accretion 10%, mergers 0%); these recoveries are overlapping and non-additive

Table 3. L3-residualized permutation importance, mid-mass growth (n = 3,430).

Feature	Raw imp.	L3-resid imp.	95% CI	Retention
		Internal class		
$\log M_{\rm gas}$	0.145	$+\mathbf{0.027}$	$[+0.017, +0.038]$	19%
All other internal	≤ 0.002	≈ 0	spans zero	absorbed
		Halo class		
$f_* = M_*/M_{\rm sub}$	0.037	$+\mathbf{0.031}$	$[+0.021, +0.044]$	84%
$\log \lambda$ (spin)	0.017	+0.007	$[+0.002, +0.013]$	41%
$\log V_\sigma$ (veldisp)	···	+0.005	$[+0.001, +0.011]$	···
$\log M_{\rm sub}$	0.031	≈ 0	spans zero	absorbed

Table 4. Assembly ablation map, mid-mass growth (n = 3,430).

Ablated family	Features removed	Geom baseline	Int marg	Recovery	Halo marg	Halo rec.
None (full L3)	···	0.185	0.135^* $[0.095, 0.184]$	···	0.083^*	···
Mergers	$n_{\rm merg}$, $n_{\rm maj}$, $t_{\rm last\,maj}$	0.183	0.135^* $[0.097, 0.183]$	$+\mathbf{0.000}$	0.081^*	−0.002
Long accretion	$\Delta \log M_{sl12}$, $\Delta \log M_{sl16}$	0.170	0.149^* $[0.108, 0.193]$	$+\mathbf{0.014}$	0.095^*	+0.012
Peak/halfmass	peak mass ratio, $t_{1/2}$	0.160	0.161^* $[0.122, 0.208]$	$+\mathbf{0.026}$	0.107^*	+0.024

NOTE—Recovery $= \Delta R^2_{\rm marg,int}$ = ablated marginal − full L3 marginal.

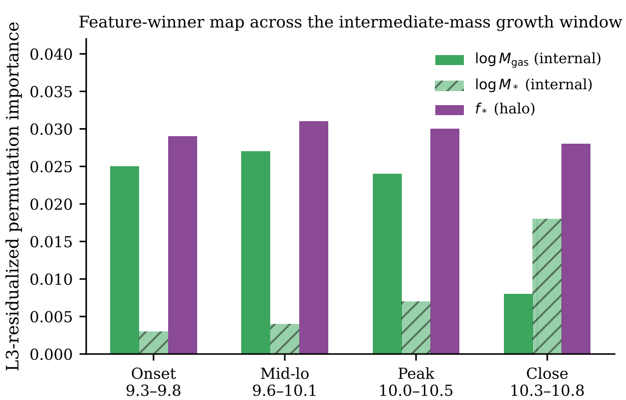

Figure 4. Feature-winner map across the intermediate-mass growth regime. L3-residualized permutation importance at four window positions. Gas mass (dark green) dominates the internal class at onset through peak; stellar mass (hatched) is the sole surviving internal feature at the close. f_* (purple) is the stable halo carrier throughout.

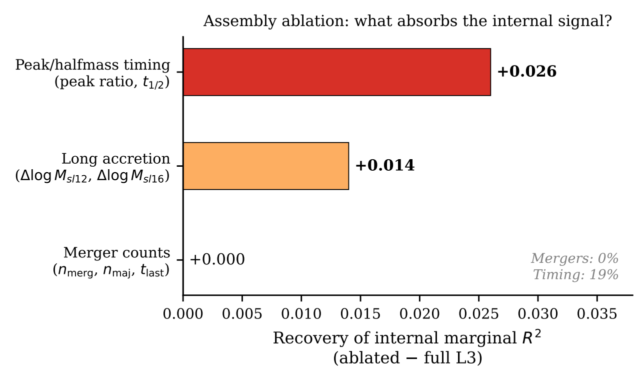

Figure 5. Assembly ablation map. Internal class marginal R^2 recovery when each gravity-feature family is removed from the L3 assembly-history: merger features (recovery +0.000), late accretion (+0.014), peak/halfmass timing (+0.026). Assembly timing is the strongest gravitational absorber of the internal signal; merger counts carry zero.

because the assembly features are collinear, and the bulk of the absorbed signal ($\approx$ 71%) is not resolved to any specific assembly-history feature in this decomposition. Extending the ablation to the early-accretion band (1–2 Gyr lookback, $\Delta \log M_{\rm halo}$ over 4 and 8 SubLink steps) does not localize the remainder: the recent-accretion pair recovers +0.009 ($\approx$ 7% of the full-L3 internal marginal) and each feature alone $\leq$ 7%, with no single assembly feature exceeding the half-mass epoch's $\approx$ 16%. Because the 13 assembly features are strongly collinear, these recoveries overlap rather than add, so the absorbed signal does not concentrate in any one feature. We therefore interpret the unresolved $\approx$ 71% as distributed, correlated assembly-history structure spread across the manifold — a decomposition limit of the present 13-feature L3 control, not a single missing absorber and not a claim of fundamental irreducibility (and, as a stronger check, a flexible non-linear assembly-history baseline does not absorb the residual gas signal either; §3.9).

The gravitational clock most correlated with baryonic growth information is *when* the halo assembled most of its mass and whether it has been gaining or losing mass since its peak, not how many mergers occurred.

3.7.3. *The internal channel is not mediated by stellar efficiency*

We add f_* directly to the L3 assembly-history baseline and re-run the assembly-history control. Adding f_* raises the baseline from $R^2 = 0.185$ to 0.251 — stellar efficiency carries $+0.066$ R^2 beyond the 13 assembly features.

Internal's marginal drops from 0.135 to 0.086 but remains significant, with a CI entirely above zero. Halo's marginal drops to 0.017 and becomes non-significant: halo's surviving L3 signal is entirely mediated through stellar-to-halo fraction. Under L3+f_*, internal leads halo 5-to-1 (0.086 vs. 0.017), compared to 1.6-to-1 under L3 alone — adding one baryonic efficiency feature exposes a qualitative difference in what the two classes are tracking.

3.8. *The Nature of the Residual Internal Signal*

What kind of information does the surviving gas signal carry? Four targeted tests on the TNG SubLink sample, using the same five-fold ridge cross-validation and shared bootstrap, characterize it (Figure 6).

First, the signal is not merely the present star-forming state. Adding the early-epoch star-formation rate, specific star-formation rate, and stellar mass to the L3 control, the gas-mass marginal for stellar growth remains positive: $+0.022\,[+0.015, +0.029]$ over 9.55–10.55 dex and $+0.069\,[+0.053, +0.086]$ at lower mass. Once gas is included the current star-formation rate adds essentially nothing, so the carrier is the gas reservoir rather than the instantaneous conversion rate.

Second, the signal is gas *amount*, not gas-use efficiency. Replacing gas amount with the depletion time $t_{\rm dep} = M_{\rm gas}/{\rm SFR}$ (which is independent of stellar mass), gas amount adds beyond $t_{\rm dep}$ ($+0.077$ at low mass, $+0.027$ over the interval) while $t_{\rm dep}$ adds nothing once gas amount is controlled. The predictive content is the amount of available fuel, not how efficiently it is currently being consumed.

Third, the reservoir is strongly shaped by halo assembly history but not exhausted by it. The L3 baseline plus stellar mass predicts $\log M_{\rm gas}$ at $R^2 \simeq 0.8$ over this mass range — dominated by halo mass and assembly timing, with stellar mass adding essentially nothing beyond L3, so this is assembly/halo-shaped gas rather than a stellar-mass artifact — yet the gas component orthogonal to that prediction still predicts future growth (residual-gas marginal $+0.077$ at low mass, $+0.039$ over the interval). Assembly history substantially determines the gas reservoir, but a predictive baryonic residual remains.

Fourth, near the upper boundary the informative predictor changes. Using catalog-level black-hole proxies — black-hole mass and accretion rate, taken as state indicators rather than direct feedback-energy measurements — black-hole/quenching state carries the growth signal at the upper transition (10.55–10.75 dex; BH-mass marginal $+0.070\,[+0.016, +0.115]$), where the gas marginal is not individually significant, and it carries the quenching signal at intermediate mass (BH-mass AUC marginal $+0.032\,[+0.025, +0.040]$), where gas does not. The high-mass bins are underpowered and the population there is largely quenched by $z = 0$ ($\gtrsim 95\%$ above 10.75 dex); this is a change in which proxy is informative, not a disappearance of the gas channel.

3.9. *Robustness*

The ordering internal > halo > env for mid-mass growth is preserved in 27/27 LOO jackknife configurations; the same holds for mid-mass quenching and for all three targets in the whole-sample analysis, so no single simulation drives any verdict. Replacing $\log M_{\rm sub}$ with $\log V_{\rm max}$ in the L1 control raises the baseline ($R^2 = 0.175$ vs. 0.004) but preserves the ordering: internal $+0.137^*\,[+0.092, +0.179]$ vs. halo $+0.075^*\,[+0.034, +0.116]$, a ratio of $\approx 1.8\times$ compared to $1.3\times$ under $M_{\rm sub}$. The lead is robust to the choice of halo structural variable.

The residual internal signal is also not an artifact of the linear ridge model. Repeating the mid-mass ($9.55 \le \log M_*/M_\odot \le 10.55$) L3-controlled stellar-growth comparison with non-linear regressors — fitting the 13-feature assembly-history baseline *itself* with the same random-forest or histogram-gradient-boosting model, so the baseline is free to capture non-linear assembly-history structure — the internal-family marginal remains positive ($+0.120\,[+0.102, +0.132]$ random forest, $+0.119\,[+0.100, +0.135]$ gradient boosting; $+0.139\,[+0.119, +0.153]$ for ridge), and gas mass remains the leading internal feature in all three by permutation importance. A more flexible, non-linear assembly-history baseline therefore does not absorb the gas signal. We adopt ridge for the main analysis for transparency and paired-bootstrap stability, not because the result depends on linearity.

Table 5. Effect of adding f_* to L3 baseline, mid-mass growth.

Class	L3 marg	L3+f_* marg	95% CI	Verdict
internal	0.135*	**0.086***	[+0.043, +0.133]	**SURVIVES**
halo	0.083*	0.017	[−0.030, +0.064]	**ABSORBED**
env	0.024 (ns)	0.013 (ns)	[−0.032, +0.060]	ns

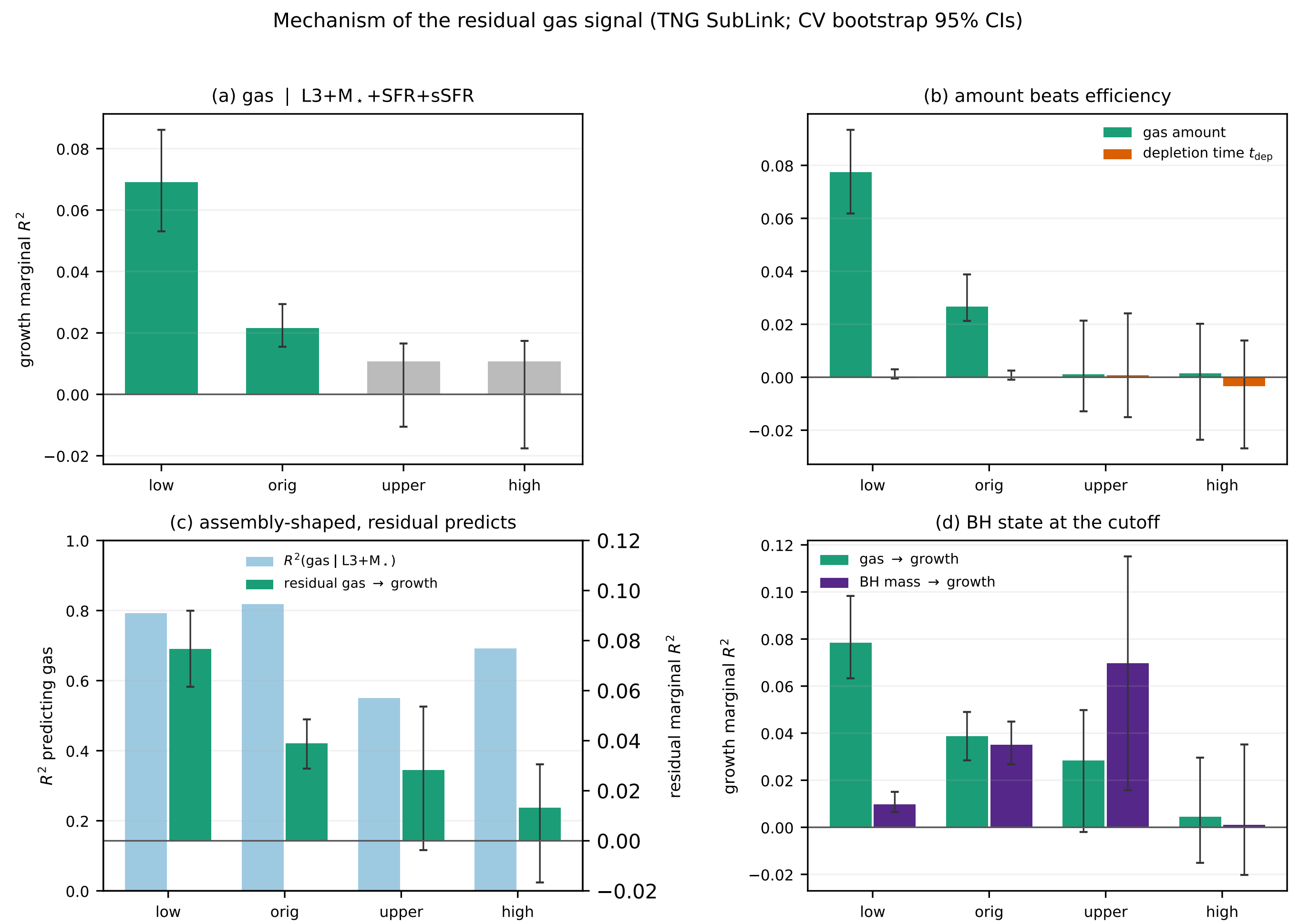

Figure 6. Mechanism of the residual gas signal in TNG (SubLink trees), by stellar-mass regime: low (9.0–9.55), *orig* (9.55–10.55), upper transition (10.55–10.75), and high (> 10.75 dex). Error bars are shared paired-bootstrap 95% confidence intervals throughout. **(a)** The gas-mass marginal for future stellar growth survives joint control for the early star-formation rate, specific star-formation rate, *and* stellar mass on top of the L3 assembly-history baseline — significant at low and intermediate mass (green), not individually significant above the cutoff (grey). **(b)** The carrier is gas *amount*, not efficiency: gas mass adds growth information beyond the depletion time $t_{\rm dep} = M_{\rm gas}/{\rm SFR}$, whereas $t_{\rm dep}$ adds essentially nothing once gas amount is included. **(c)** The reservoir is strongly shaped by assembly history — the L3+$M_\star$ baseline predicts $\log M_{\rm gas}$ at $R^2 \simeq 0.8$ (light bars, left axis) — yet the orthogonal residual gas still predicts growth (green, right axis): assembly-shaped, but not exhausted. **(d)** Near the upper boundary the informative predictor changes hands from gas to black-hole mass — used as a black-hole/quenching *state* proxy, not a direct feedback-energy measurement: the gas→growth marginal fades while the BH-mass→growth marginal peaks at the upper transition.

3.10. *Cross-Simulation Comparison at a Matched (L1) Control*

The mechanistic results of Sections 3.4–3.7 are based on TNG SubLink. To test family robustness, we run the L1 sliding-window mass scan on SIMBA spatial (Figure 7), restricting to the growth and quenching targets.

At a matched L1 control, the two simulations differ sharply in the growth channel: SIMBA's L1 growth-channel internal-family marginal does not reach robust significance (scan peak +0.080, 95% CI spanning zero), and any SIMBA growth signal is weak and fragmented, whereas TNG's is a clean +0.245 over the same control. The SIMBA L1 internal–halo paired gap for growth is correspondingly significant at only two isolated mass centres.

The quenching channel is the opposite. At the same L1 control SIMBA's internal-family quenching marginal is large and broadly significant (peak marginal AUC +0.141* at 10.15 dex, significant from 9.65 to 10.75 dex; Table 6; halo also significant, peak +0.127*), whereas TNG's quenching marginal is small (+0.060* at L1). The same-control L1 contrast in the quenching channel is thus ∼2.4× in SIMBA's favour.

The TNG quenching marginal is weak at every control level: peak L1 AUC +0.060* at 9.85 dex, tightening further under L3, confirming that TNG's beyond-gravity signal does not lie in the quenching channel (Figure 7, lower-left). We compare the two simulations only at the shared L1 control; we deliberately do *not* set SIMBA's L1 quenching marginal against TNG's more conservative L3 marginal, as that would confound the target-channel question with the depth of the gravitational control.

At a matched L1 control, then, the two feedback models express their residual beyond-gravity internal information through different target channels: in TNG it concentrates in stellar growth (the upper-bounded intermediate-mass regime of Section 3.4), whereas in SIMBA it concentrates in quenching (broadly significant L1 band, peak marginal AUC +0.141), with SIMBA's growth signal weak and fragmented.

We stress the scope of this contrast. The TNG branch supports the full mechanistic decomposition — static position versus accretion history versus full assembly history — via merger-tree matching. The SIMBA branch supports only a same-control L1 target-channel contrast: SubLink-equivalent merger trees are unavailable for SIMBA at CAMELS resolution, so we make *no* claim of an L3-controlled residual in SIMBA. The SIMBA result should be read as evidence that the target channel of the beyond-gravity signal differs across feedback models at a matched static control, not as a matched mechanistic replication of the TNG assembly-history decomposition.

4. DISCUSSION

4.1. *Interpretation: From Fuel-Limited to Quenching-Limited Growth*

Across an upper-bounded intermediate-mass regime (significant under L3 up to $\log M_*/M_\odot \simeq 10.55$ in this suite), a galaxy's baryonic state at $z \approx 0.77$ carries *predictive* information about its $z = 0$ stellar growth that is not fully captured by its gravitational trajectory — even given static position, accretion rate over multiple lookback windows, peak mass, half-mass assembly epoch, merger counts, and major-merger timing. The literal result is a *predictive* transition: at low-to-intermediate mass the residual gas reservoir is the dominant beyond-gravity predictor of future growth, while near and above the upper boundary black-hole/quenching state becomes the more informative channel. This is consistent with a transition from a fuel-limited to a quenching-limited regime, but our measurements are predictive, not causal.

We do not interpret the lower edge as a physical onset: below ~ 9.55 dex the recovery of the gas signal is limited by catalog-floor and resolution effects (§3.4), so the apparent low-mass onset is a recovery edge rather than a physical transition. Whether a genuine low-mass change exists (e.g. a greater role for environment via ram pressure, tidal stripping, or filamentary gas supply (A. Dekel et al. 2009)) cannot be settled at this resolution.

The upper boundary near ~ 10.55 dex is physically meaningful: there the predictive structure shifts to black-hole/quenching state (traced by catalog-level black-hole proxies, §3.8) and the population is largely quenched by $z = 0$, so the regime is bounded above by the onset of quenching rather than by assembly history alone. In the intermediate regime below it, the current gas *amount* carries predictive information that the 13 tested gravitational descriptors do not fully recover. The upper boundary is defined within the CAMELS CV setup and may shift modestly with volume and resolution (see §4.6).

The regime has a carrier structure within it (Section 3.7): gas mass dominates the surviving internal signal from the onset through the peak, but shifts to accumulated stellar mass near the close. This is consistent with evidence that cold-gas content is a strong predictor of near-future star formation at intermediate mass (A. Saintonge et al. 2017; J. J. Davies et al. 2020), and with a picture where, toward higher mass, the gas-to-growth link is increasingly absorbed by gravitational history while the galaxy's position in the quenching trajectory — reflected in accumulated stellar mass (S. Genel et al. 2018) — retains an independent predictive contribution.

Observational prediction. This residual gas-reservoir signal is not merely a simulation result — it is a quantitative prediction testable on real galaxy surveys. The prediction is: in observed galaxies at $9.55 \lesssim \log M_* \lesssim 10.55$ (roughly $z \lesssim 0.1$), cold-gas mass should retain marginal predictive power for stellar growth ~ 1–2 Gyr later, above and beyond what halo-mass proxies (weak-lensing mass, group membership, velocity dis-

Table 6. SIMBA L1 quenching marginal AUC, selected windows.

Centre (dex)	n	Int marg AUC	Sig	Halo marg AUC	Sig
9.65	1,225	+0.068	*	+0.045	ns
9.85	1,098	+0.111	*	+0.111	*
10.15	855	**+0.141**	*	+0.127	*
10.25	825	+0.125	*	+0.096	*
10.35–10.45	···	+0.123–+0.117	*	+0.100–+0.083	*
10.75	550	+0.086	*	+0.083	*
$\geq$ 10.85	···	ns		ns	

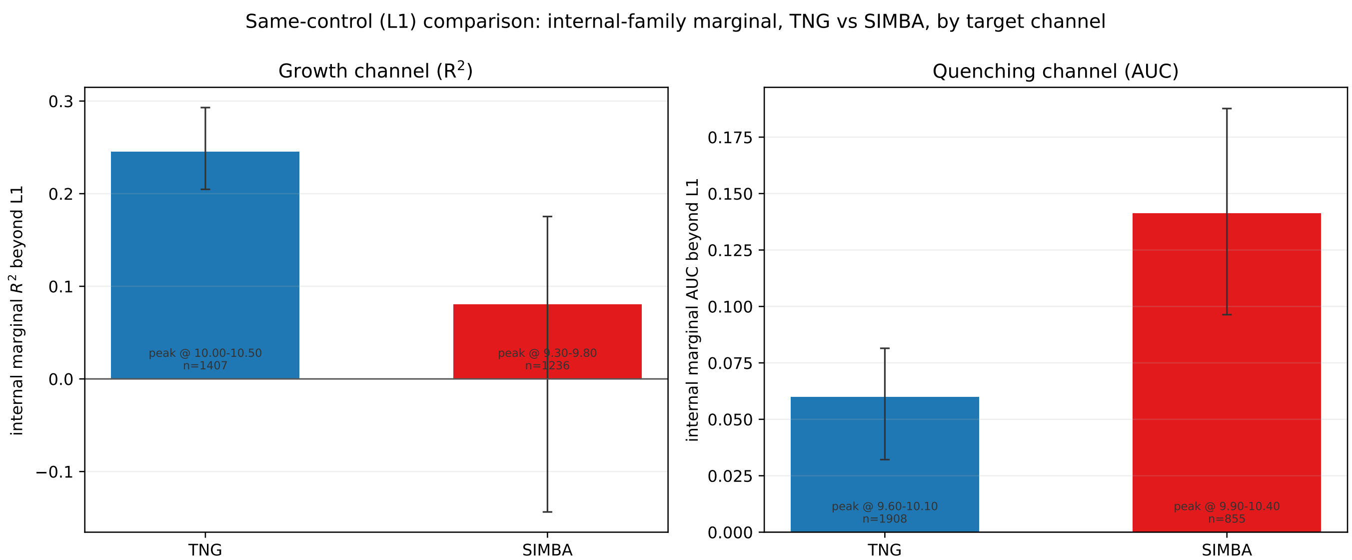

Figure 7. Cross-simulation comparison at a matched L1 (static-halo) control: internal-family marginal for TNG (blue) and SIMBA (red), by target channel. *Left*: future stellar growth (marginal R^2 beyond L1). TNG carries a clean internal signal ($+0.245\,[+0.205, +0.293]$), whereas SIMBA's is weak and not robustly significant (peak $+0.080$, 95% CI spanning zero). *Right*: quenching (marginal AUC beyond L1). The ordering reverses — SIMBA's internal quenching marginal is large ($+0.141^{*}\,[+0.096, +0.188]$) while TNG's is small ($+0.060^{*}\,[+0.032, +0.081]$), an L1-vs-L1 amplitude ratio of $\sim$2.4$\times$ in SIMBA's favour. Error bars are shared paired-bootstrap 95% confidence intervals; the peak mass band and sample size are annotated on each bar. Tree-based assembly (L3) controls are unavailable for SIMBA at CAMELS resolution, so the comparison is made only at the shared L1 control and no SIMBA L3 residual is claimed.

persion) already predict. The xGASS and xCOLD GASS surveys (B. Catinella et al. 2018; A. Saintonge et al. 2017) observe total cold-gas content ($\mathrm{HI} + \mathrm{H_2}$) for star-forming galaxies in precisely this stellar-mass range, with group-membership and halo-mass proxies available from companion catalogs. A marginal-predictive battery run on xGASS/xCOLD GASS — with halo-mass proxies as the assembly-history control and cold-gas mass as the test predictor — would constitute an observational *proxy* test of the regime. Because the simulation-side predictor is *total* subhalo gas mass and the CAMELS target is the $\sim$ 6–7 Gyr growth from $z \approx 0.77$ to $z = 0$, whereas the surveys measure cold ($\mathrm{HI} + \mathrm{H_2}$) gas and nearer-term growth at low redshift, this should be read as a test of whether the accessible cold-gas reservoir traces the *same residual gas-reservoir channel* — not a one-to-one replication of the exact CAMELS total-gas, $\sim$ 6–7 Gyr prediction. If the signal survives in observed galaxies, it would establish that the predictive contribution of gas content relative to halo assembly is not an artifact of the CAMELS feedback prescriptions.

4.2. *What the Result Does Not Mean*

Three misreadings of the result should be avoided. It is not anti-gravitational: the assembly-history baseline accounts for roughly half the internal family's signal in this regime (L3 retention ≈ 43% from an L1 baseline near 100%), so halo assembly history remains the dominant information source and the internal signal is a residual beyond it. It does not mean that only baryonic information matters here: halo structure also survives L3 in the regime, albeit in fewer mass bins and with a smaller marginal. Nor is it universal: the residual is confined to the intermediate regime — recovery-limited at low mass and underpowered at high mass — and the whole-sample tie of Section 3.1 is real.

4.3. *Why Assembly Timing is the Main Gravitational Competitor*

The assembly ablation result (Section 3.7) constrains which part of the halo assembly-history baseline most overlaps with the internal predictive signal. Individually, peak mass ratio and half-mass assembly snapshot recover 19% of the absorbed internal advantage, late-lookback accretion 10%, and merger counts 0%. Because the assembly features are collinear, these per-component recoveries are overlapping and non-additive: no single decomposed channel accounts for the bulk of the absorbed signal. This is a collinearity/decomposition limitation of the L3 control rather than evidence that the absorbed signal is individually irreducible.

Peak mass ratio and half-mass assembly epoch together encode the galaxy's gravitational *maturity* in the sense of C. Lacey & S. Cole (1993) — when it assembled most of its mass and whether it has been on a declining trajectory since (R. H. Wechsler et al. 2006; Y.-Y. Mao et al. 2018). A galaxy that peaked early and has been stripped carries different predictive content from one still near its peak mass, even at the same present-day halo mass.

This timing information partially *overlaps* with the baryonic signal because a galaxy near its mass peak is more likely to have a well-stocked gas reservoir at the relevant epoch (A. Saintonge et al. 2017). The overlap is partial and predictive, not a causal decomposition — which is precisely why the internal signal survives after ablating those features.

Merger counts do not participate in this overlap. The number of mergers experienced, or when the last major merger occurred, does not substantially predict the current gas mass or the future growth rate, net of the other assembly descriptors. The residual internal signal in the mid-mass regime is merger-count-independent.

4.4. *The f_* Boundary*

Adding f_* to the assembly-history baseline absorbs halo's surviving L3 marginal entirely while leaving internal's intact. This is a sharp result: internal carries baryonic growth information that is orthogonal to both the halo assembly-history baseline and the integrated efficiency of star formation. Halo's surviving gravity-controlled signal, by contrast, was tracking formation efficiency all along.

The physical implication is that the baryonic channel cannot be reduced to "the galaxy happens to be efficient at forming stars." Whatever the gas-mass signal captures — current fueling conditions, feedback state, gas phase — it is not recoverable from the stellar-to-halo ratio record even in combination with 13 assembly features.

4.5. *Why the Family Contrast Is Meaningful, and What It Does Not Yet Prove*

The SIMBA result is not a robustness check for the TNG growth result; it is a different measurement. SIMBA shows that a beyond-gravity baryonic signal exists in SIMBA too, but it is concentrated in quenching rather than stellar growth. At this matched L1 control, the simplest reading is a feedback-model difference in target-channel expression (scoped to L1, not a mechanistic claim): TNG's prescription leaves more independent information in the fuel-to-growth channel, while SIMBA's stronger wind quenching leaves more in the quenching channel.

In TNG, the mechanistic decomposition is clean: we have merger-tree assembly-history control escalating from static position through full assembly history, assembly ablations, and the f_* augmentation test. In SIMBA, we have a target-channel comparison at L1 only. The SIMBA result is best understood as evidence for a family-level difference in which baryonic outcome carries the beyond-gravity signal, not as a parallel mechanistic decomposition. Resolving whether the SIMBA quenching signal survives an equivalent L3 assembly-history control requires SIMBA SubLink-equivalent trees, which are not currently available at CAMELS resolution.

4.6. *Caveats*

Several limitations apply to these results. Even at its best, the internal predictor family explains only $R^2 \approx 0.31$ of mid-mass growth variance, leaving ≈ 69% unexplained. This residual likely reflects stochastic processes not recoverable from snapshot features, finite resolution suppressing small-scale ISM physics, the compression of internal state into six scalar features, and genuine physical stochasticity.

The CAMELS CV suite uses $25\,h^{-1}$ Mpc boxes with 256^3 particles. At the low-mass end the lower edge of the regime is a floor-encoding / resolution-limited recovery boundary (§3.4), not a physical onset: a small tail of unresolved objects whose gas feature sits at the catalog floor destabilizes the low-mass fit. We do not have a higher-resolution CAMELS run with which to test convergence, so we report the 9.55-dex onset as a recovery edge and the upper boundary near 10.55 dex as the physically meaningful edge. The small box also coarsely samples large-scale modes, so the environmental null — environment carries no marginal beyond L1 in any window — is specific to the CAMELS CV volume and the environment features used here, and we do not claim it as a general result. The CV suite holds astrophysical parameters fixed, so results are valid for the specific TNG and SIMBA prescriptions at CAMELS resolution; the effect of varying feedback strength is not explored. Finally, direct AGN feedback-energy fields are not present in the CAMELS group catalogs used here, so the high-mass transition is characterized through black-hole state proxies (black-hole mass and accretion rate), not a feedback-energy measurement. Relatedly, the loaded group-catalog products provide total subhalo gas mass but no phase-separated cold or star-forming gas reservoir (no cold-, neutral-, or molecular-gas mass field is present), so our gas predictor is total gas. The observational prediction is correspondingly framed in terms of total gas-reservoir content and should be tested with cold-gas surveys; isolating the cold or star-forming phase on the simulation side would require particle-level or value-added data beyond the loaded catalogs. At the relevant stellar masses the predictive signal is therefore interpreted as a gas-*reservoir* signal, but the present catalog-level analysis cannot decompose it into cold, star-forming, and hot phases; the proposed cold-gas test accordingly probes whether the observable cold-gas reservoir traces the same residual channel rather than the exact simulation variable.

On the SIMBA side, SubLink-equivalent merger trees are not available, so L3 assembly-history control cannot be applied. The SIMBA quenching result (Section 3.10) is at L1 only. All findings are correlational throughout: predictive advantage means a family carries information about the target, not that it causes the outcome. The 13 L3 assembly-history features were selected for physical breadth — covering static position, multiple accretion-rate lookbacks, peak-mass epoch, half-mass assembly, and merger history — rather than optimized for predictive importance against the internal target. A feature-selected or higher-dimensional assembly-history control could absorb a larger share of the internal signal, though the robustness of the surviving signal in the permutation-importance test (gas mass) and the f_* boundary check suggests this result is not dominated by L3 feature-set sensitivity. Finally, because SubLink matching requires a matched central descendant, the analysis excludes galaxies that merge or are disrupted between $z = 0.77$ and $z = 0$. These objects may have different gas reservoirs and growth histories from the retained population, so all results are conditional on the matched central-galaxy sample; we do not claim a specific direction for how their inclusion would change the internal marginal.

5. CONCLUSIONS

We have compared three predictor families — internal galaxy properties, halo structural properties, and environmental measures — as predictors of stellar growth, quenching, and halo growth in CAMELS CV IllustrisTNG ($n = 7{,}362$) and SIMBA ($n = 4{,}203$) central galaxies, with escalating controls for halo structural position and assembly history. The main findings are:

1. **No feature family leads consistently**: every cell in the 3×3 score matrix is a statistical tie in both TNG and SIMBA. The overall tie pattern is simulation-robust but does not describe the full mass-regime structure.

2. **An upper-bounded intermediate-mass gas-reservoir regime**: under the full 13-feature assembly-history control, internal galaxy state retains significant marginal predictive power for stellar growth up to $\log M_*/M_\odot \simeq 10.55$ (11 overlapping significant windows over the catalog-recovered interval 9.55–10.55 dex, peak internal marginal $R^2 = 0.161^*$, 43% retention at the fixed mid-mass window). The lower edge is a floor-encoding/resolution-limited recovery boundary, not a physical onset; outside this regime feature-family advantages are largely assembly-history-mediated.

3. **Internal leads halo across 9.55–10.75 dex**: the shared-bootstrap paired gap (internal − halo) under L3 is positive and significant at every window in this range (13 overlapping windows, +0.052–+0.089, all CIs positive), extending two windows beyond the 9.55–10.55 dex internal interval. The gap is flat — it does not peak at a single mass.

4. **Gas mass is the dominant surviving internal carrier**: the only internal feature retaining significant permutation importance after L3 residualization. Its assembly-history-residualized importance is 19% of its raw importance, broadly consistent with the family-level retention.

5. **Assembly timing, not mergers, is the main absorbing gravity channel**: removing peak mass ratio and half-mass assembly snap from L3 recovers 19% of the absorbed internal marginal; removing merger counts recovers 0%. The mid-mass internal signal is merger-count-independent.

6. **The internal channel is not mediated by stellar efficiency**: adding f_* to the L3 assembly-history absorbs halo's surviving marginal entirely ($0.083^* \to 0.017$ ns) but leaves internal's intact ($0.135^* \to 0.086^*$ $[+0.043, +0.133]$). Internal carries baryonic growth information orthogonal to both gravitational assembly history and galaxy formation efficiency.

7. **A beyond-gravity baryonic signal appears in both simulation families, but its target-channel expression differs**: in TNG it is expressed through stellar growth (11-window L3 interval; paired gap over 13 windows, +0.052–+0.089); in SIMBA, at a matched L1 control, the stronger signal appears in quenching (10+ window L1 band, peak marginal AUC = 0.141^*), while the SIMBA growth signal is fragmented and not robustly significant. The TNG branch supports the full mechanistic decomposition; the SIMBA branch supports a target-channel contrast. Whether the SIMBA quenching signal survives full assembly-history control requires merger-tree equivalents not yet available for SIMBA at CAMELS resolution.

A detailed accounting of what is and is not claimed by these results — including explicit statements on the absence of causal claims, universal mass thresholds, observational inference, and cross-simulation replication — is given in Appendix B. Directions for follow-up (SIMBA merger-tree control, volume and resolution tests, and redshift evolution of the regime) are collected in Appendix C.

ACKNOWLEDGMENTS

The author thanks the open-source scientific Python community for the libraries underlying this work: NumPy, SciPy, pandas, scikit-learn, matplotlib, and tqdm. The CAMELS simulations were produced by the CAMELS collaboration and are publicly available through the Flatiron Institute data portal. CAMELS IllustrisTNG uses the IllustrisTNG galaxy-formation model developed by the IllustrisTNG collaboration; CAMELS SIMBA uses the SIMBA model developed by R. Davé and collaborators.

DATA AND CODE AVAILABILITY

The CAMELS simulation data used in this work are publicly available at https://camels.readthedocs.io. All analysis code, figure-generation scripts, and the compiled paper are available at https://github.com/kunalb541/Camels.

APPENDIX

A. ADDITIONAL MECHANISM AND ROBUSTNESS DIAGNOSTICS

The compact mechanism summary in the main text (Figure 6) and the cross-simulation comparison (Figure 7) are distilled from a broader set of diagnostics, reproduced here in full so that the supporting evidence is visible rather than asserted. Unless noted, all reuse the paper pipeline (five-fold ridge cross-validated R^2 for growth, logistic cross-validated AUC for quenching, shared paired-bootstrap 95% confidence intervals) on the TNG SubLink sample.

The gas reservoir is separated from the present star-forming state in Figure 8: controlling jointly for the early star-formation rate, specific star-formation rate, and stellar mass on top of L3, the gas-mass marginal for future growth remains $+0.022\,[+0.015, +0.029]$ over 9.55–10.55 dex and $+0.069\,[+0.053, +0.086]$ at lower mass. The raw gas-*fraction* marginal is inflated by its stellar-mass denominator (+0.121); the leakage-free gas-mass quantity is the +0.022 value quoted throughout.

Figure 9 contrasts gas amount with the depletion time $t_{\rm dep} = M_{\rm gas}/{\rm SFR}$ (constructed to be independent of stellar mass). Gas amount adds beyond $t_{\rm dep}$ (+0.077 at low mass, +0.027 over the interval), whereas $t_{\rm dep}$ adds essentially nothing once gas amount is included ($\simeq -0.000$): the predictive content is the amount of available fuel, not the instantaneous conversion efficiency.

The reservoir's encoding by assembly history is quantified in Figure 10: the L3+$M_\star$ baseline predicts $\log M_{\rm gas}$ at $R^2 = 0.793$ (low) and 0.818 (interval) — largely set by halo mass and assembly timing, with stellar mass adding essentially nothing beyond L3 — yet the gas component orthogonal to that prediction still predicts future growth (+0.077 low, +0.039 interval). The reservoir is assembly-shaped but not exhausted.

Figure 11 follows the black-hole proxies (mass and accretion rate, used as state indicators rather than feedback-energy measurements). The BH-mass growth marginal peaks at the upper transition ($+0.070\,[+0.016, +0.115]$, where the

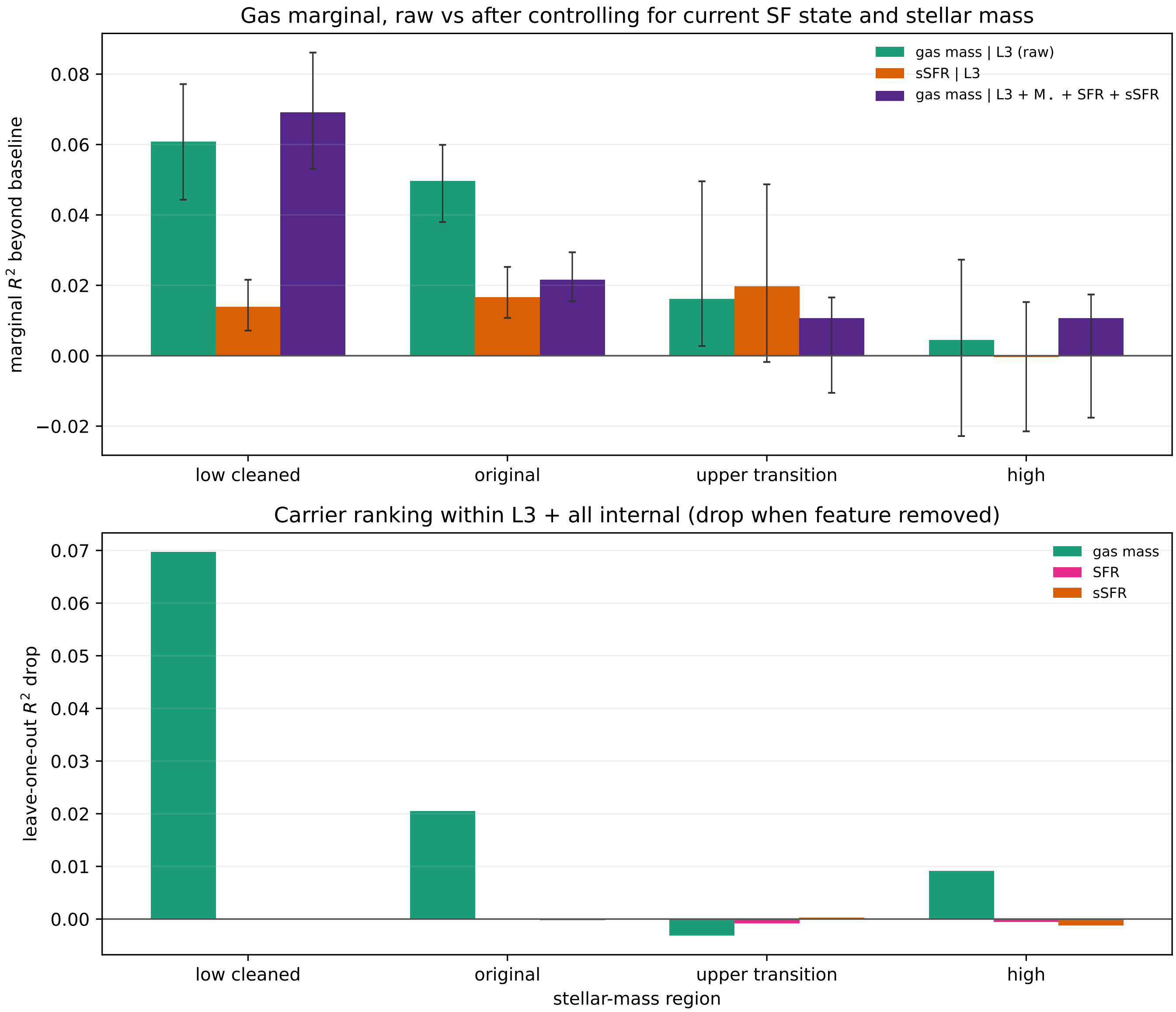

Figure 8. Gas-reservoir versus current-star-formation discriminator. After controlling for the early SFR, sSFR, and stellar mass, the gas-mass marginal for future growth stays positive (+0.022 over the interval, +0.069 at low mass); the raw gas-fraction marginal (+0.121) is inflated by its stellar-mass denominator and is shown only to disclose that inflation.

gas marginal is not individually significant), and the BH-mass quenching marginal is significant at intermediate mass (AUC $+0.032\,[+0.025, +0.040]$); above 10.75 dex the population is $\gtrsim 95\%$ quenched by $z = 0$.

A coarsened-exact-matching test of the gas→growth effect at fixed stellar mass, halo mass and assembly history is shown in Figure 12. A positive effect is recovered where matching is feasible — low mass, ATT $= +0.138\,[+0.105, +0.172]$ dex at borderline covariate balance ($|\mathrm{SMD}| = 0.14$; the quoted interval reflects bootstrap scatter only and does not include matching/balance uncertainty) — while the intermediate and upper regimes are matching-infeasible (insufficient high-gas/low-gas overlap at fixed covariates). We treat this as supportive robustness only in the low-mass region where matching is feasible — not a global causal estimate, and not the basis for the main intermediate-mass claim, where matching is infeasible — which is why it appears here rather than in the main text.

Finally, Figure 13 establishes that the lower edge is not a physical threshold. Below 9.55 dex the full-sample gas marginal is destabilised by a few objects whose gas feature is pinned at the catalog floor (estimate +0.005 with a meaningless interval $[-0.279, +0.010]$); deleting that tail gives +0.061, and, crucially, *winsorizing* the floor-encoded

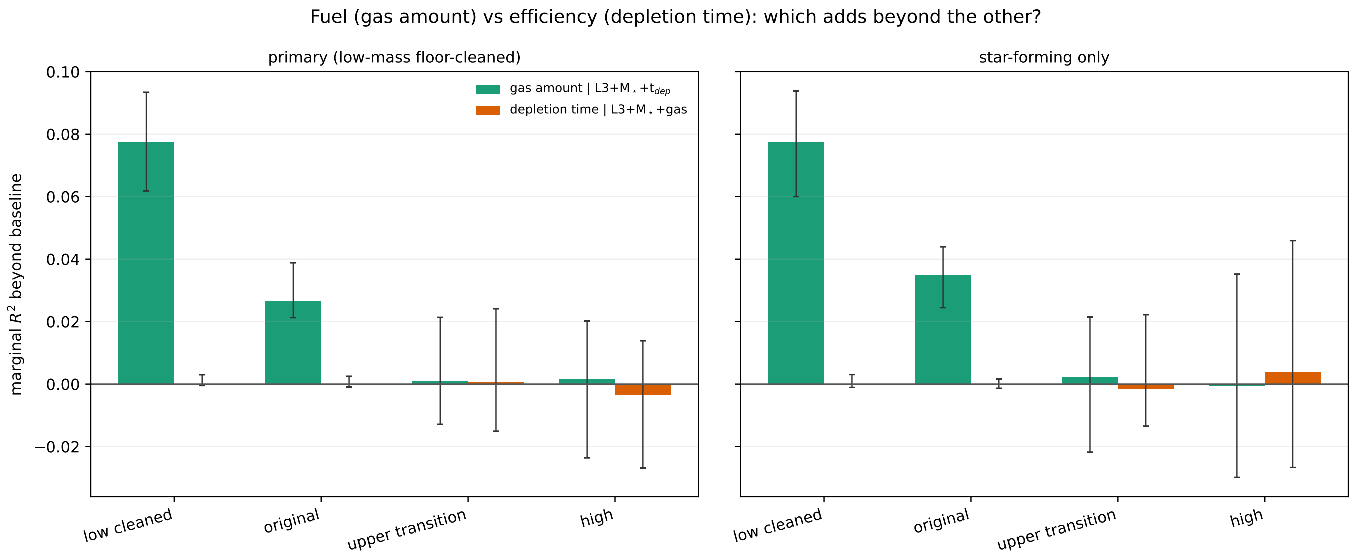

Figure 9. Gas amount versus depletion time. Gas amount carries growth information beyond $t_{\rm dep} = M_{\rm gas}/{\rm SFR}$, while $t_{\rm dep}$ adds essentially nothing once gas amount is controlled — the signal is fuel amount, not conversion efficiency.

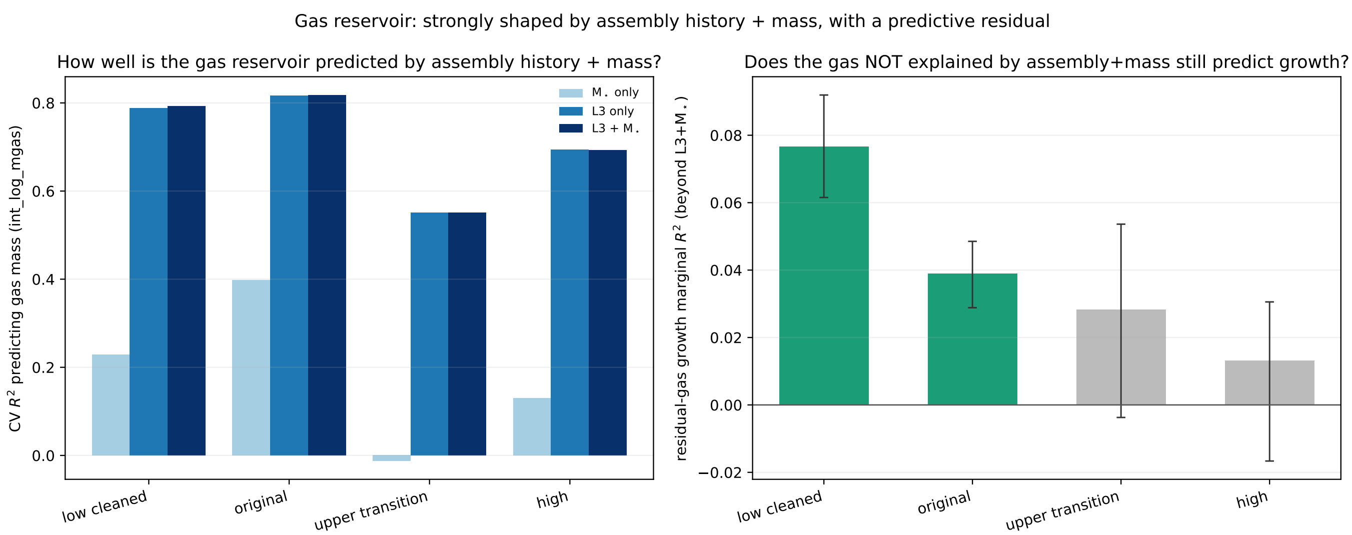

Figure 10. Assembly encoding of the gas reservoir. The L3+$M_\star$ baseline predicts the reservoir at $R^2 \simeq 0.8$, but the orthogonal residual gas retains a positive growth marginal (+0.077 low, +0.039 interval): the reservoir is shaped by halo assembly history without being fully determined by it.

feature without removing any object recovers +0.057 — matching the onset band (+0.061). The lower edge is therefore a floor-encoding / resolution-limited measurement edge.

Finally, Figure 14 shows the population profiles requested for the boundary characterization: across the upper transition ($\log M_\star \sim 10.55$) the median specific star-formation rate drops, the quenched fraction rises, and the median black-hole mass increases, supporting the upper bound as a quenching-state transition. These are population diagnostics, not inputs to the predictive model.

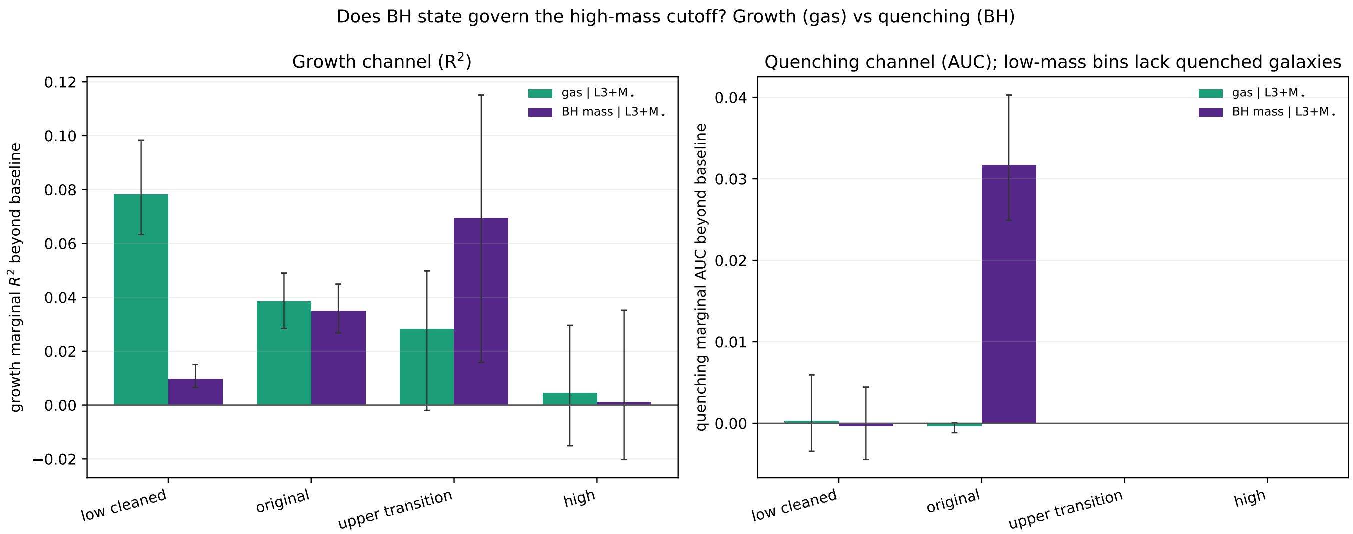

Figure 11. Black-hole state at the upper transition. Black-hole mass (a state proxy, not a feedback-energy measurement) carries the growth signal at the upper transition and the quenching signal at intermediate mass — precisely where the gas marginal is not individually significant.

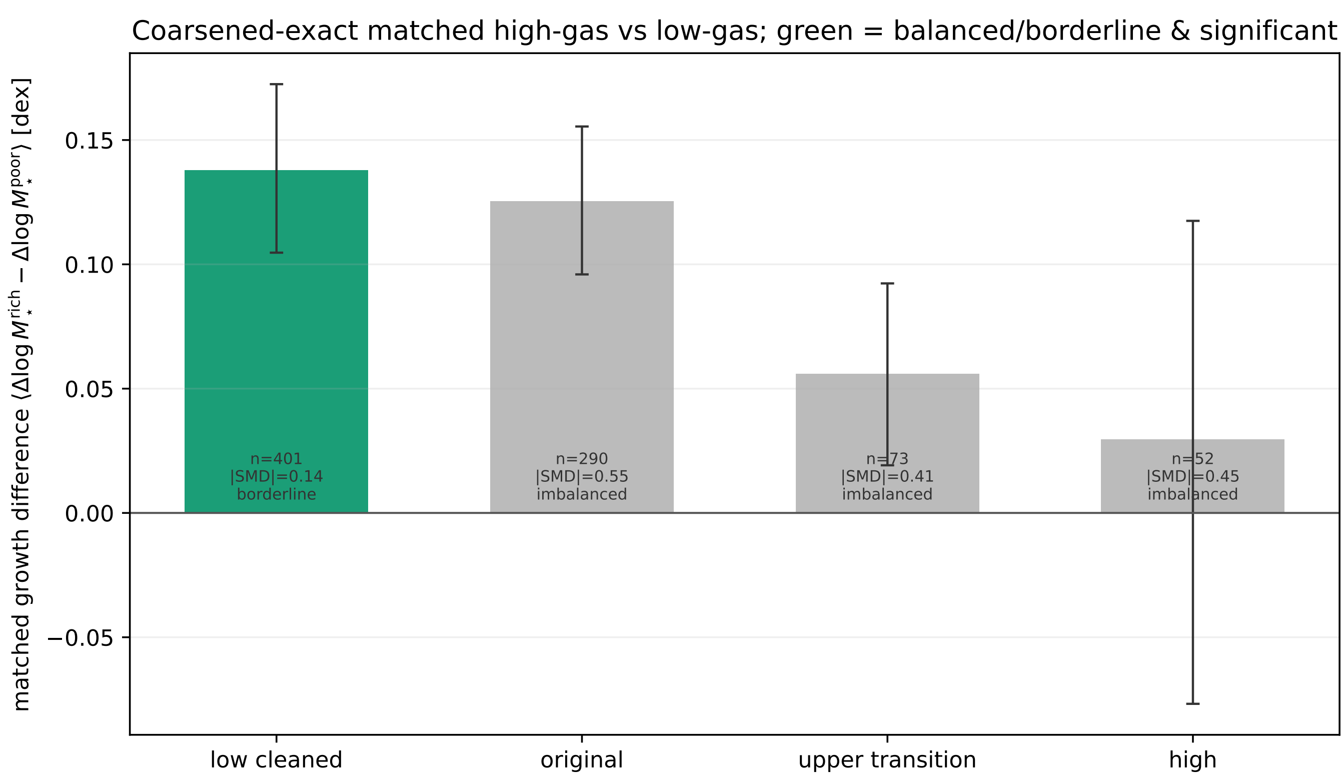

Figure 12. Coarsened-exact-matched gas→growth test. At low mass, where high-gas and low-gas galaxies can be matched on stellar mass, halo mass and assembly history, the matched effect is ATT = +0.138 dex (borderline balance, |SMD| = 0.14); the higher-mass regimes lack sufficient overlap to match. Supportive robustness, not a global causal estimate.

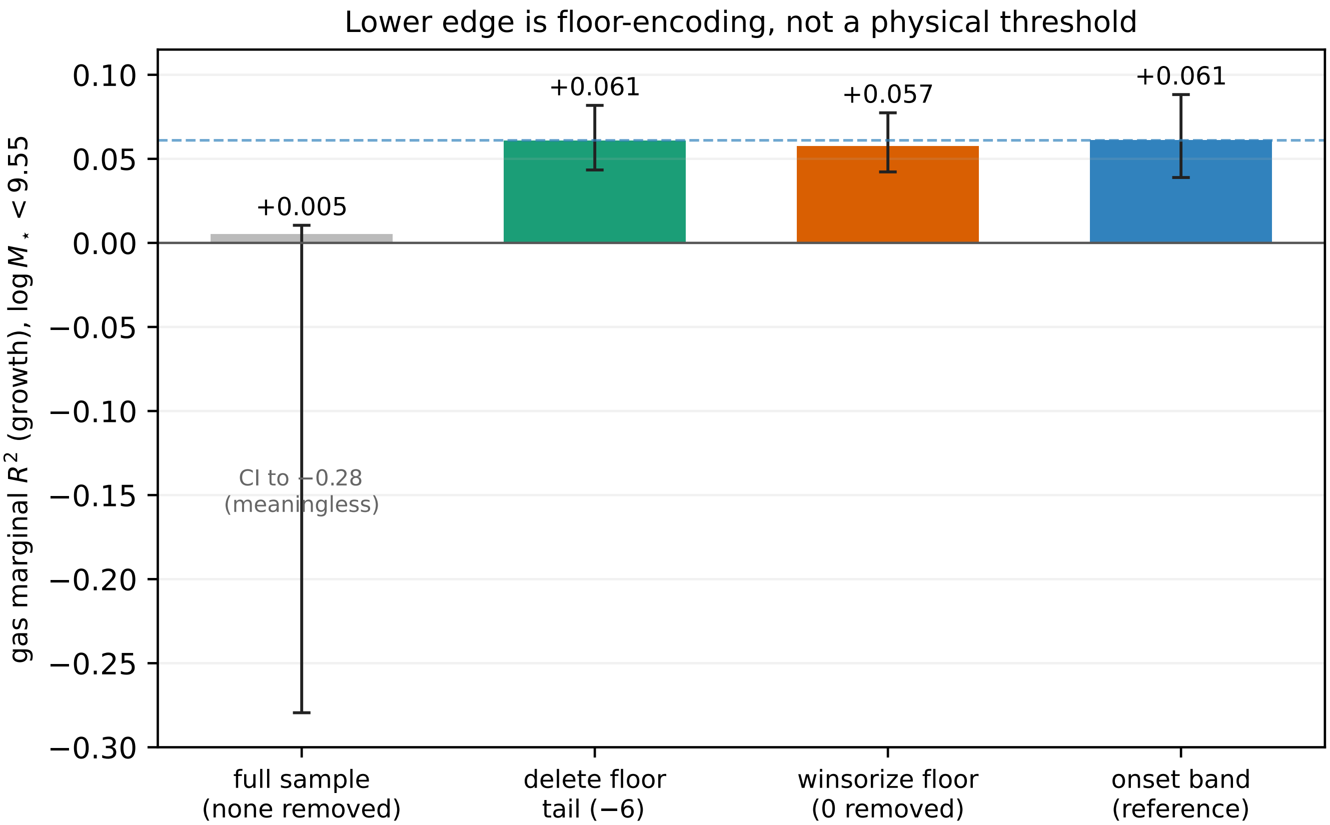

Figure 13. Lower-edge floor encoding. Deleting (−6 objects) or winsorizing (32 objects, none removed) the floor-pinned gas feature recovers the same gas marginal (+0.061 / +0.057) as the onset reference band (+0.061), whereas the raw full-sample estimate (+0.005) carries a meaningless interval reaching −0.28. The lower edge is a floor-encoding / resolution-limited measurement edge, not a physical threshold.

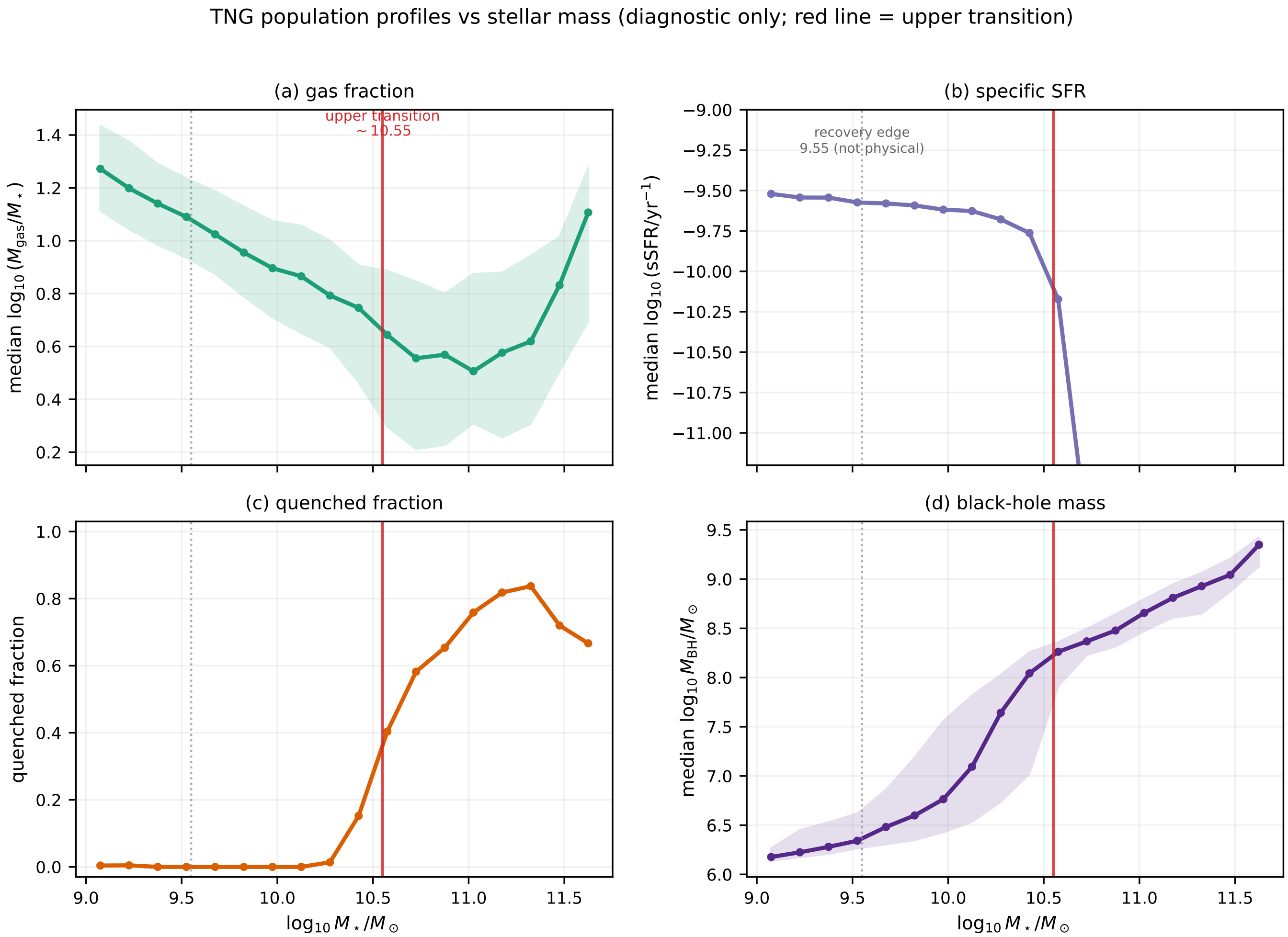

Figure 14. Boundary population profiles in TNG as a function of stellar mass (0.15-dex bins): **(a)** median gas fraction $\log_{10}(M_{\rm gas}/M_\star)$, **(b)** median specific star-formation rate, **(c)** quenched fraction ($\log \mathrm{sSFR} < -11$), and **(d)** median black-hole mass (a state proxy, not a feedback-energy measurement). The red line marks the upper transition near $\log M_\star \simeq 10.55$; the dotted line marks the 9.55 recovery edge (a floor/resolution-limited measurement edge, not a physical threshold). Across the upper transition the specific star-formation rate drops, the quenched fraction rises, and the black-hole mass increases together, consistent with the onset of black-hole/quenching-associated suppression of star formation. *These profiles are population diagnostics, not part of the predictive model.*

B. SCOPE OF CLAIMS

Earned in this paper.

- *No single feature family is globally best across the tested targets.* Across the whole-sample target matrix, environment, internal, and halo families are all competitive; the ranking changes with both target and mass regime.
- *An upper-bounded mid-mass growth regime exists in TNG in which internal galaxy state adds predictive information beyond the tested gravitational prescription.* In the TNG SubLink branch, internal features retain positive marginal R^2 for future stellar growth after conditioning on the full 13-feature assembly-history control.
- *That internal advantage is regime-specific, not global.* The beyond-gravity signal is concentrated in a bounded intermediate-mass interval rather than extending across the full sample.
- *The internal-minus-halo gap is positive across that interval under the full TNG gravity control.* Internal state is not merely surviving control; it remains ahead of halo structure across the interval.
- *Gas mass is the dominant surviving internal carrier.* Within the internal family, gas mass is the main load-bearing predictor after the full gravity control.
- *Halo assembly timing is the main absorbing gravitational channel.* Peak-mass ratio and half-mass assembly timing recover substantially more absorbed internal signal than merger-count variables.
- *Merger counts do not explain the internal growth signal.* Removing merger-count features in the ablation recovers essentially none of the absorbed internal marginal.
- *The surviving internal signal is not mediated by stellar-to-halo fraction f_*.* Adding f_* to the baseline absorbs halo's residual but leaves the internal signal positive and significant.
- *The matching-method confound is real and quantitatively important.* Spatial matching inflates the geometry baseline by 3.7× relative to merger-tree matching; mechanistic interpretation should rest on the SubLink/TNG branch.
- *SIMBA supports a cross-family channel contrast, but not a matched mechanistic replication.* TNG expresses the beyond-gravity signal primarily in growth; SIMBA expresses it primarily in quenching.

Not earned in this paper.

- *A universal internal-state superiority claim.* The paper shows a bounded, regime-specific TNG result, not a general dominance of internal state across galaxy evolution targets.
- *Universal mass boundaries.* The quoted interval edges are earned within the CAMELS setup and should not be read as universal physical thresholds independent of volume, resolution, or feedback model; the lower edge in particular is resolution-limited.
- *A causal claim.* These are predictive and decomposition results. The paper shows information beyond the tested gravitational prescription, not causal independence from gravity.
- *A statement that gravity is unimportant.* The full assembly-history baseline explains a large share of the signal; the surviving internal component is a residual beyond that baseline, not a replacement for it.
- *A clean mechanistic replication in SIMBA.* Without tree-based control for SIMBA, the SIMBA result is comparative support, not a matched mechanistic replication.
- *A resolution- or volume-independent result.* The paper does not establish that the interval location or width is unchanged under larger boxes or higher resolution.
- *A direct observational claim.* This is a simulation result on feature families and predictive structure, not an observational inference about real galaxies.
- *Proof that gas physics is independent of halo history.* The paper shows residual predictive information after the tested gravity controls, not absolute independence between gas content and assembly.

C. FUTURE DIRECTIONS

The two most direct extensions are:

SIMBA merger-tree control. The SIMBA branch uses only spatial matching, which inflates halo baselines by 3.7× and prevents a direct L3-equivalent test. Constructing SubLink-like trees for SIMBA-CAMELS would enable a fully symmetric comparison and determine whether the SIMBA quenching signal survives a 13-feature assembly-history control.

Volume and resolution. Results come from $25\,h^{-1}$ Mpc boxes with 256^3 particles. Running the same analysis on larger CAMELS extensions will determine whether the predictive interval shifts or broadens at higher mass.

Two further simulation directions are worth pursuing: (*i*) repeating the sliding-window scan at $z \approx 0.5$ and 1.0 to track whether the regime evolves as assembly history accumulates; (*ii*) tracking net gas inflows and outflows along individual TNG trajectories to identify whether the gas-mass signal reflects accretion or feedback-driven depletion. The observational proxy test of the upper-bounded gas-reservoir regime via xGASS/xCOLD GASS (B. Catinella et al. 2018; A. Saintonge et al. 2017) is discussed and formulated as a prediction in Section 4.

D. SUPPORTING TABLES

Table 7. Predictive scores by stellar mass regime, TNG SubLink. Bold = WINNER at raw (uncontrolled) level; the assembly-history-controlled gap for mid-mass growth is +0.052–+0.089 across 13 overlapping windows (Table 2).

Target	Mass bin	n	env	halo	internal	Verdict
Growth R^2	Low (< 9.5)	2,902	0.166	0.160	0.040	TIE
	Mid (9.5–10.5)	3,430	0.083	0.235	**0.307**	**WINNER: internal**
	High (≥ 10.5)	1,030	0.068	0.107	0.118	TIE
Quenching AUC	Low (< 9.5)	2,902	0.645	0.704	0.681	TIE
	Mid (9.5–10.5)	3,430	0.791	0.807	0.818	TIE
	High (≥ 10.5)	1,030	0.791	0.792	0.735	TIE

Table 8. Whole-sample predictive scores by feature family and simulation. TNG SubLink: $n = 7{,}362$; SIMBA spatial: $n = 4{,}203$. Bold = highest score in column.

Target	Family	TNG SubLink	SIMBA Spatial	Verdict
Growth R^2	env	0.133	0.274	TIE
	halo	**0.257**	0.342	TIE
	internal	0.247	**0.354**	TIE
Quenching AUC	env	0.846	0.717	TIE
	halo	0.862	0.737	TIE
	internal	**0.865**	**0.743**	TIE
Halo growth R^2	env	0.008	0.151	TIE
	halo	**0.015**	**0.158**	TIE
	internal	0.006	0.144	TIE

Table 9. Assembly-history baseline by matching method.

Matching	Family	Baseline R^2
SubLink	TNG	$0.059\,[0.051, 0.072]$
Spatial	TNG	0.220
Spatial	SIMBA	0.273

Table 10. L1 marginal R^2 by window centre, selected positions (full scan in Figure 2). The L1 control removes only static halo position (3 features); a positive marginal means that family adds predictive power *beyond* structural position alone. Internal is significant from 9.55–10.65 dex (13 windows); halo enters three windows earlier. The L3 assembly-history-controlled equivalent is in Table 11. $*$ denotes 95% CI lower bound > 0.

Centre (dex)	n	Int marg	Sig	Halo marg	Sig
9.25–9.45	⋯	< 0, wide CI	ns	+0.053–+0.072	*
9.55	2,393	+**0.149**	*	+0.078	*
9.65–10.45	⋯	+0.153–+0.244	*	+0.082–+0.171	*
10.25	1,407	+**0.245**	*	+0.171	*
10.55–10.65	⋯	+0.182–+0.111	*	+0.107–+0.060	*
10.75	836	+0.083	ns	+0.031	ns
$\geq$ 10.85	⋯	ns		ns	

Table 11. L3 internal marginal R^2 by window centre (selected). $*$ denotes 95% CI lower bound > 0.

Centre (dex)	L1 int marg	L3 int marg	L3 Sig	L3 halo marg	L3 halo Sig
9.25–9.45	< 0	< 0	ns	ns	ns
9.55	+0.149	+**0.110**	*	+0.039	ns
9.65–9.75	+0.153–+0.164	+0.110–+0.117	*	ns	
9.85	+0.180	+0.132	*	+0.061	*
10.05–10.45	+0.175–+0.225	+0.135–+0.161	*	+0.063–+0.072	ns–*
10.35	+0.244	**0.161**	*	+0.072	*
10.55	+0.182	+**0.106**	*	+0.054	ns
10.65	+0.111	+0.056	ns	ns	
$\geq$ 10.75	ns	ns		ns	

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